JEREMIAH RICHARD

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Objective

Mechanical engineering graduate and current M.S. Robotics student at Northwestern University seeking engineering opportunities in robotics, systems integration, automation, embedded systems, mechanical design, manufacturing, product development, or applied technology. Background includes CAD, structural analysis, CNC machining, mechatronics, Python, machine learning, and project leadership, with a strong interest in practical problem-solving and hands-on engineering design.

Education

Northwestern University | Evanston, Illinois
Master of Science in Robotics | Expected December 2027
Current Graduate Focus: Robot manipulation, kinematics, embedded robotics, ROS 2, Linux-based robotics development, simulation, and statistical machine learning/data analysis.
MSR Hackathon Exposure: Git, ROS 2, RRT path planning, Python robot control, RGB-D sensing, and basic robot vision/calibration concepts.
Planned Coursework (not yet completed): Mechatronics, C++ robotics software development, mobile-robot control, sensor fusion, SLAM, sensing/navigation, and machine learning for robotics.

Hardin-Simmons University | Abilene, Texas
Bachelor of Science in Mechanical Engineering | Graduated Spring 2026
Math Minor
GPA: 3.76 | Magna Cum Laude
Relevant Coursework:

- Statics, Dynamics
- Mechanics of Materials
- Thermodynamics
- Fluid Mechanics
- Circuits
- FEA (ANSYS)
- Heat Transfer
- Advanced Mechanics & Design
- Instrumentation & Controls
- Vibrations & Kinematics
- Data Science & Machine Learning
- Material Science

Technical Skills

- CAD & Simulation: SolidWorks, AutoCAD, Inventor, ANSYS, SolidWorks CAM, Fusion 360
- Programming & Data Analysis: Python, MATLAB, Jupyter Notebook, Pandas, NumPy, Matplotlib
- Machine Learning: Scikit-learn, Supervised Learning, Unsupervised Learning, Regression, Classification, Clustering, Model Evaluation
- Robotics & Software: ROS 2 (developing), CoppeliaSim, Linux/Ubuntu, Git/GitHub, VS Code
- Manufacturing: HAAS CNC Mill, 3D Printing, CNC Plasma Cutting, Basic Welding, Hand & Power Tools
- Mechatronics: Microcontrollers, Servo Motors, Sensors

Projects

- **Capstone Project – Assistive Patient Lifting System for Hippotherapy Program** — Led a senior design team in developing a personnel-assist lifting system for a university hippotherapy program, including structural analysis, safety-focused design, component selection, budgeting, scheduling, and fabrication planning. Independently learned to operate a HAAS CNC mill for project manufacturing; supported standards-conscious engineering decisions using applicable OSHA, ASME, and ADA considerations; completed a 625 lb proof test of the final structure.
- **Machine Learning Coursework Projects** — Completed supervised and unsupervised learning projects in Python, including classification, regression, and clustering analysis. Applied data cleaning, feature selection, train/test splitting, model training, and evaluation using accuracy, RMSE, R^2, and cluster interpretation.
- **Junior Design Project – Dumbbell Rack and Storage Cabinet** — Designed and constructed a dumbbell rack and cabinet for a university fitness center using welding, CNC machining, structural analysis, caster wheel load evaluation, and finite element analysis.
- **Mechatronics – Webcam-Controlled Rover** — Worked on an Arduino-based rover using MATLAB computer-vision concepts, sensors, and servo control; gained hardware/software integration and debugging experience before the project ended prior to full autonomous completion.
- **Shaft Analysis Project** — Performed static and fatigue analysis to determine maximum rated horsepower using gear force analysis, bending stress, torsional stress, von Mises stress, and safety factor evaluation.
- **Kinematics & Vibrations Project** — Built and tested a vibration apparatus integrating a spring–mass oscillator and bending beam; conducted experimental design, data collection, and vibration analysis.

- **API Development Project** — Built an animal fact API demonstrating programming logic, API structure, and problem-solving skills.
- **Python Minecraft Project** — Designed and implemented a Python-based game in Google Colab using programming fundamentals and interactive logic.

Professional Experience

- **Arca Continental Coca-Cola Southwest Beverages** – Manufacturing Engineering Intern Support (May 2025 – Aug 2025)
  - Improved water efficiency in CIP sanitation processes, supporting sustainability goals.
  - Analyzed rinse cycle performance and identified inefficiencies.
  - Translated raw data into actionable insights, balancing water reduction and food safety.
- **Best Buy** – Sales Advisor & Product Flow (May 2024 – Present)
  - Advised customers on technology products; exceeded membership and credit sales goals.
  - Managed online order fulfillment and inventory stocking efficiently.
- **Adidas** – Retail Store Associate (Jun 2022 – Aug 2022)
  - Assisted customers with product selection; top seller for multiple weeks.
- **Smoothie King** – Team Member (Jun 2021 – Aug 2021)
  - Delivered excellent customer service in a fast-paced environment.
- **Superb Cleansing** – Owner, Lawn Care & Pressure Washing (May 2019 – Aug 2020)
  - Founded and managed a successful local business serving 50+ clients.

Extracurricular & Community Activities

- Student-Athlete – NCAA Football
- 2× Academic All-American; maintained 3.72 GPA while competing at collegiate level; graduated 3.76 Magna Cum Laude.
- Developed leadership and time management skills balancing academics, athletics, and work.
- PALs Program – Mentored children ages 6–10 in a nationally recognized, structured school leadership and activities program during high school.
- Guest Educational Presenter – Delivered a live animal presentation on ball pythons to K–8 students, covering species origin, African cultural history, and responsible pet ownership.

Awards & Achievements

- **Dean's List** (Fall 2025, Spring 2024, Spring/Fall 2023)
- **President's List** (Fall 2024, Spring 2026)
- Graduated High School with 4.53 GPA (Top 2%)

References

Available upon request.